Laboratoire de Recherche en Intelligence Artificielle

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جامعة هواري بومدين للعلوم والتكنولوجيا
USTHB

DATA MINING

Course: M2 SII 2025-2026

By: Dr. N. A. Houacine

Outline

Chapter 1: Introduction

Chapter 2: Data, Dataset, Data Warehouse

Chapter 3: Exploratory Data Analysis

Chapter 4: Data Cleaning & Preprocessing

Chapter 5: Feature Engineering

Chapter 6: Supervised Machine Learning

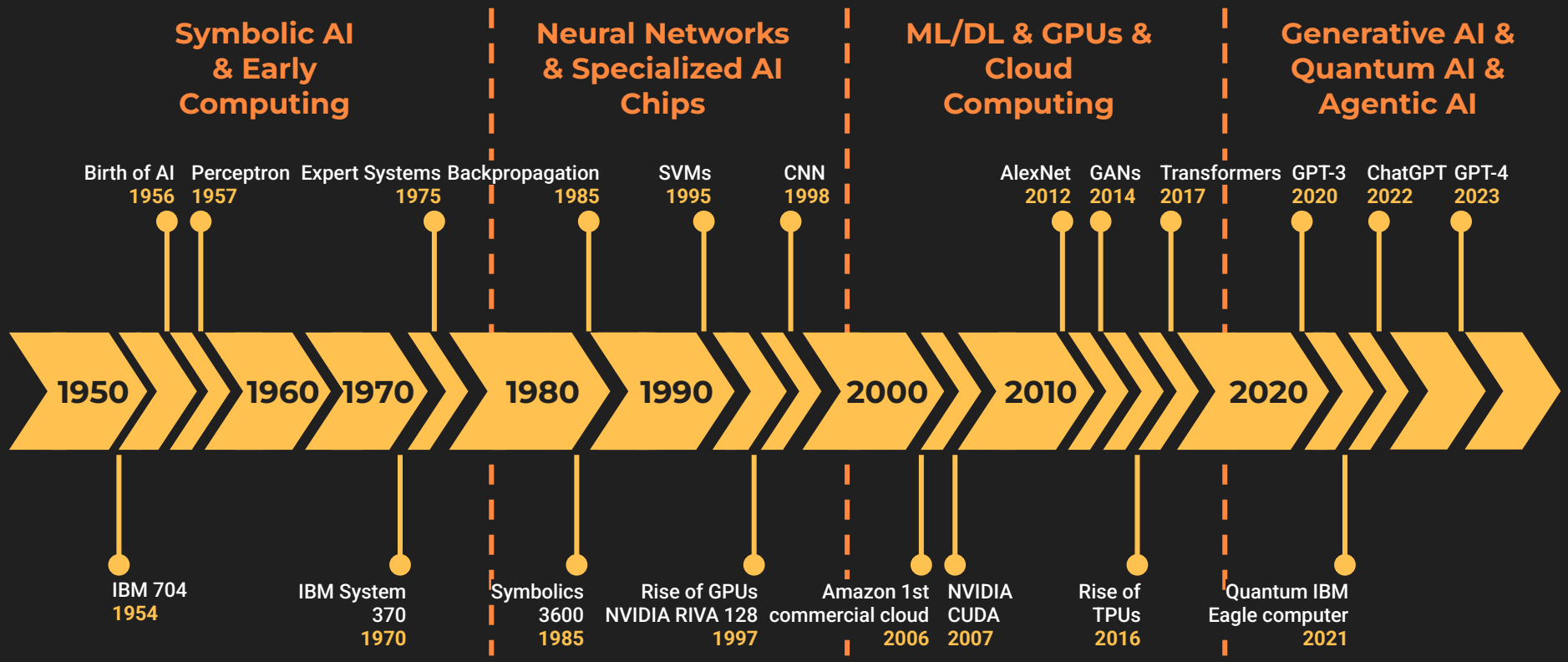
Chapter 7: Unsupervised Machine Learning

Chapter 8: Something more

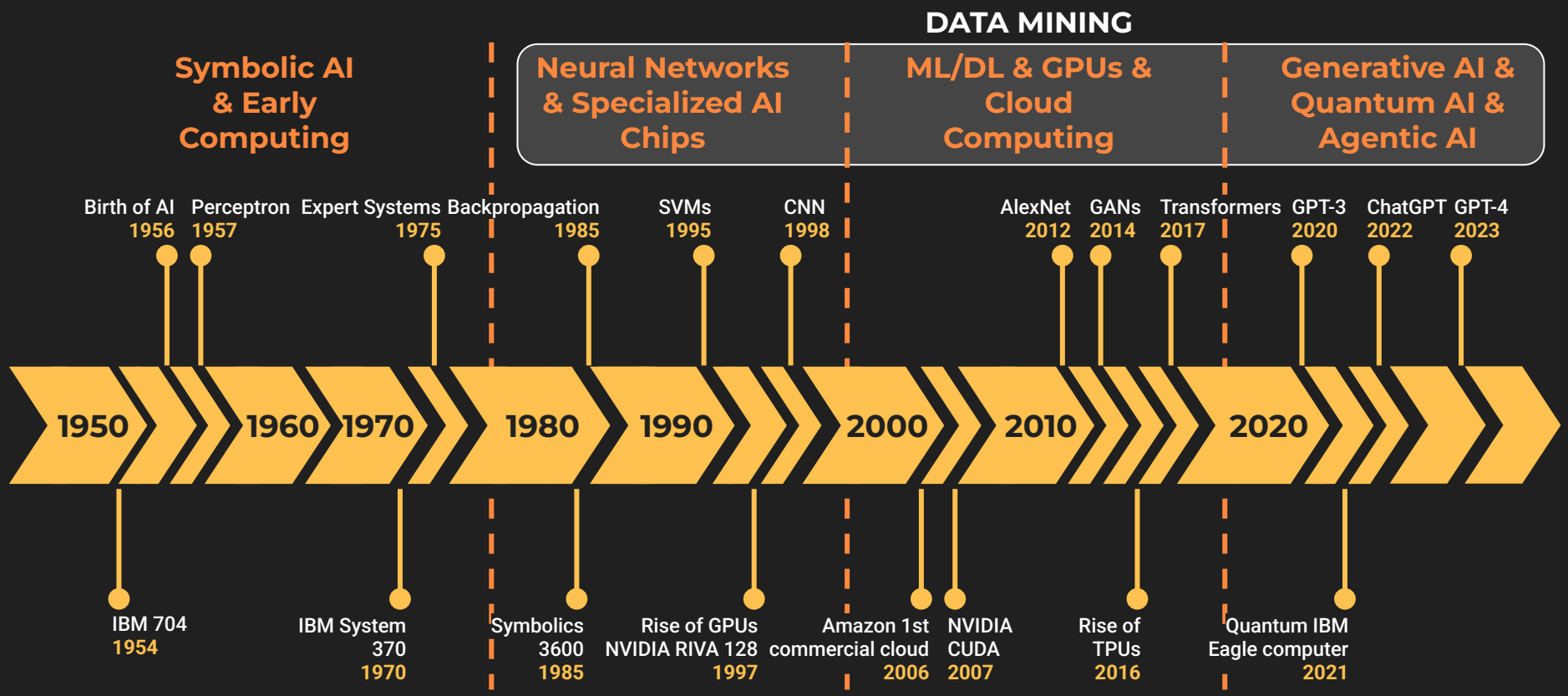
1. Introduction

AI *VS* DATA Mining? What is data mining? The knowledge discovery process?

From Symbolic AI to Modern AI



From Symbolic AI to Modern AI



The Pillars of Modern AI

Modern Artificial Intelligence, and by extension, data mining, stands on three interdependent **pillars**. Understanding these components is crucial for grasping the capabilities and limitations of AI systems.

Data

The raw material for insights. Quality, volume, and variety are paramount. Without data, there is no AI.

Algorithms

The computational logic that processes data to find patterns and make decisions. These are the "brains" of AI.

Hardware

The physical infrastructure that provides the computational power for complex algorithms to run on large datasets.

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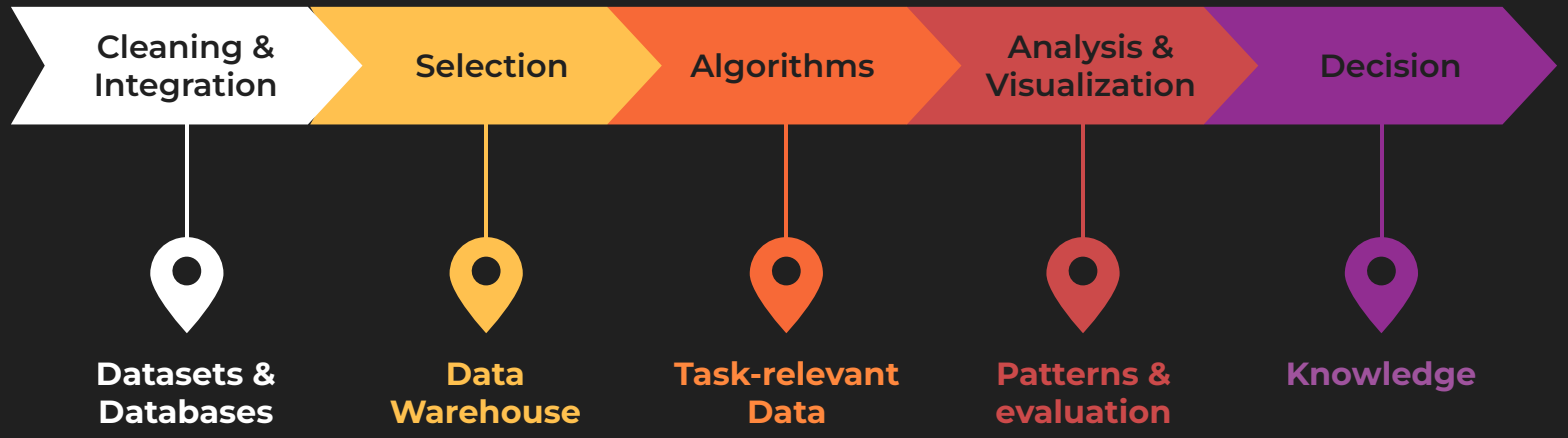
DATA MINING

What is Data Mining ?

Data mining is the overall process of discovering patterns and extracting useful insights from large datasets through the use of statistics, mathematics, and machine learning algorithms. *~IBM*

It's more than just collecting data; it's about making sense of it to drive informed decisions. Data mining acts as a bridge between raw data and actionable knowledge.

Data Mining General Process



What is Data Mining ?

The field of data mining is interdisciplinary and has evolved under various names, reflecting its diverse applications and underlying methodologies. While "data mining" is the most common term, you might encounter these synonyms:

- Knowledge Discovery in Databases (KDD),
- Knowledge Extraction,
- Data/Pattern Analysis,
- Business intelligence (BI),
- Big Data Analytics, etc.

Each term emphasizes a slightly different aspect, but all converge on the goal of extracting value from data.

Data Mining in Business Intelligence

06 Decision Making

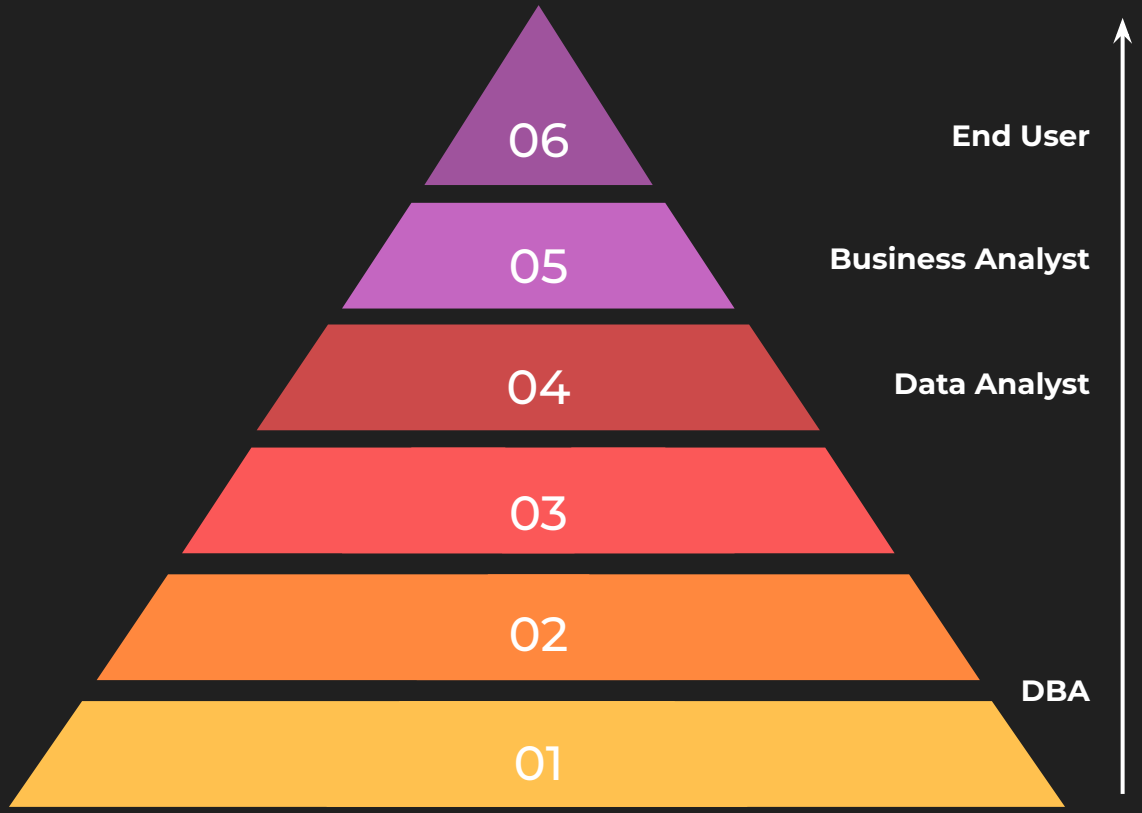
05 Data Presentation
Visualization techniques

04 Data Mining
Information Discovery

03 Data Exploration
Statistical summary, Querying, Reporting.

02 Data Preprocessing/ Integration
Dataset, Data Warehouse, ...

01 Data Sources
Paper, Files, Web docs, database systems, ...



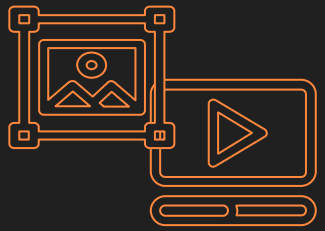
What Data can be Mined ?

Database-oriented data :

- Relational database,
- Data warehouse,
- Transactional database

Advanced datasets :

- Data streams and sensor data
- Time-series / sequence data,
- Structure data (graphs, social networks and multi-linked data)
- Heterogeneous databases
- Spatial data
- Spatiotemporal data
- Multimedia database
- Text databases



What Patterns can be Mined ?

Multidimensional description

Characterization,
Generalization,
Summarization,
Contrast data, etc.

Association and Correlation Analysis

Frequent patterns,
Association,
Extract rules,
Correlation vs. causality

Classification and Prediction

Construct models that describe and distinguish classes or concepts for future prediction.

Cluster Analysis

Unknown class labels,
Groups the data to form new categories.
Discover pattern distributions.

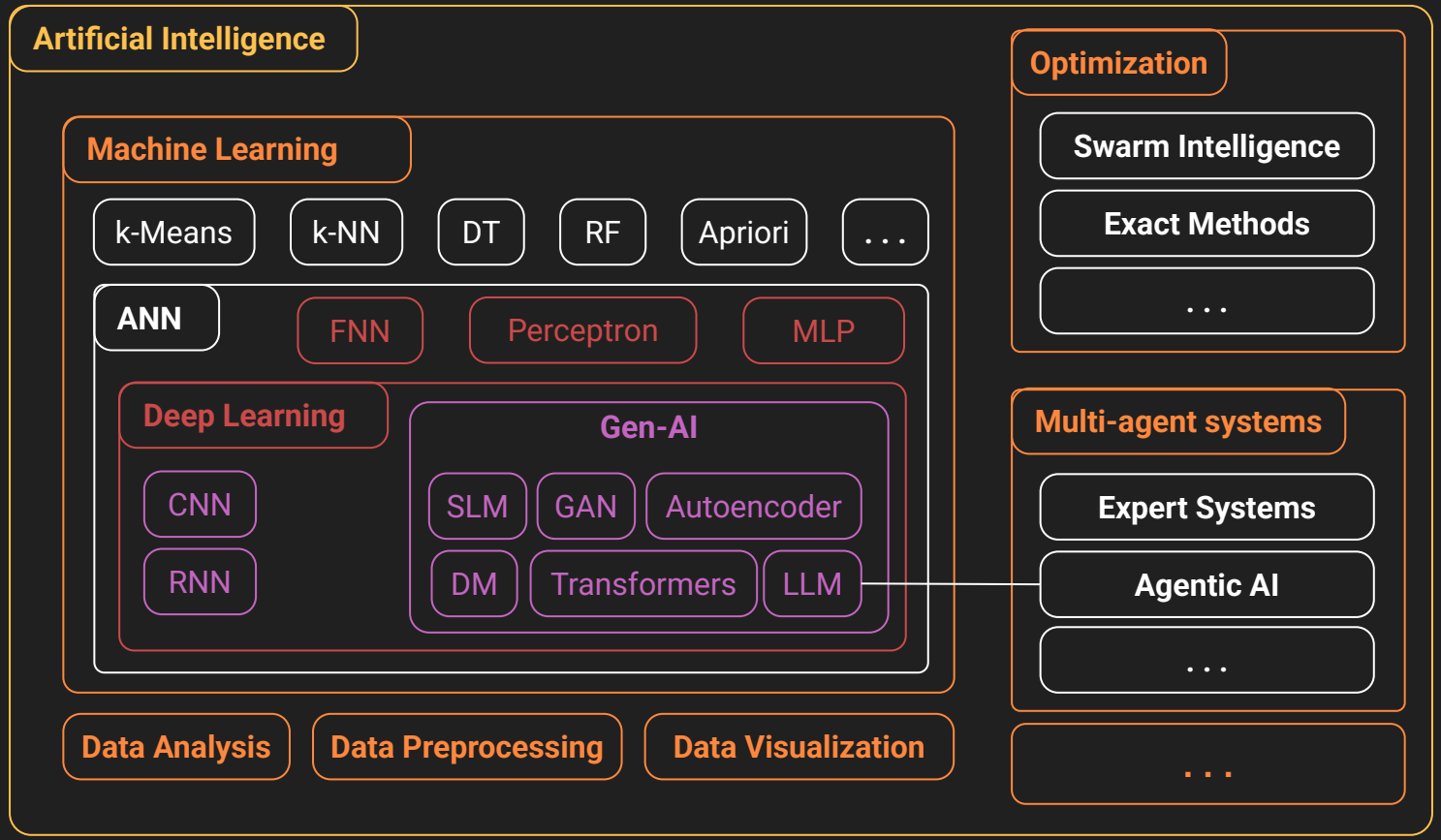
Anomaly detection

identifying rare, suspicious data points or patterns that significantly deviate from an established normal behavior.

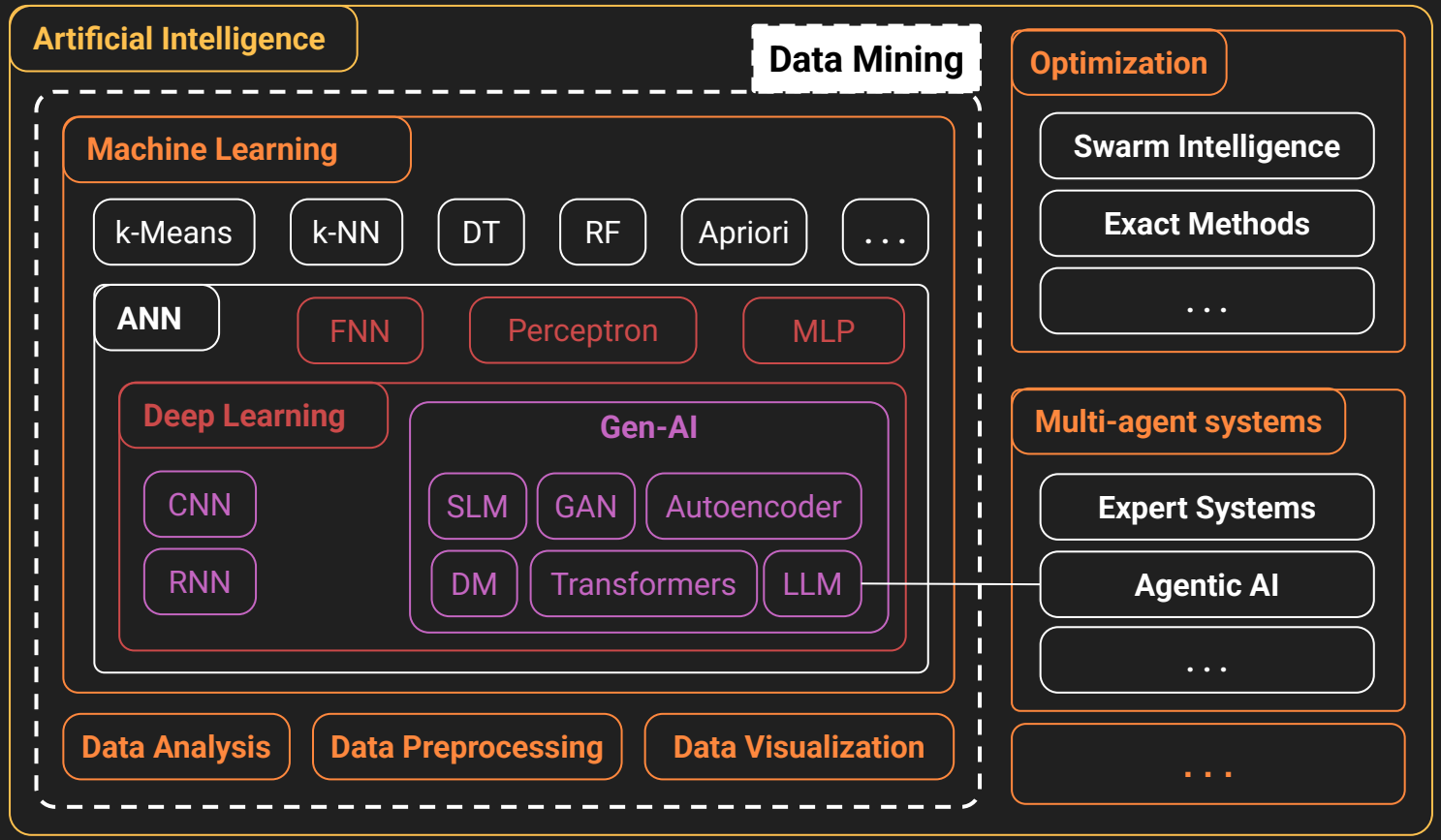
Other Analysis

Trends, sequential patterns,
Periodicity analysis,
Statistical analysis,
...etc.

Artificial Intelligence *VS* Data Mining



Artificial Intelligence *VS* Data Mining



Course structure

Chapter 2 : Data, Dataset, Data Warehouse
Datasets, Characteristics of structured data, Data Objects and attributes types. Distance measures. Dataset vs Data Warehouse

Chapter 3 : Exploratory Data Analysis
Univariate, Bivariate, and Multivariate analysis

Chapter 4 : Data Cleaning & Preprocessing
Data quality measures. Handling missing data and outliers. Dimensionality & numerosity reduction

Chapter 5 : Feature Engineering
Feature Extraction, Selection, and Scaling. Data Encoding.

Chapter 6 :Supervised Machine Learning
Classification, Regression tasks, Evaluation of the models. Splitting the datasets.

Chapter 7 : Unsupervised Machine Learning
Clustering tasks, Frequent Patterns and association rules, anomaly detection, and dimensionality reduction.

Chapter 8 : Something more
Some other concepts such as: Hyper-parameters tuning and model interpretation.

Course evaluation

Séances de Cours		
- Interactives dans les deux sens	- Recherches et présentations	- Envoi des slides à la fin de chaque séance
Evaluation continu		
Test - Annoncé à l'avance - Noté sur 10 - Plot twist	**Assiduité** - Présence, préparation, participation - Noté sur 5	**Projet TP** - Par binôme chaque séance - Rapport à la fin - Noté sur 25
EMD		
	- Tous les chapitres - Noté sur 20	